

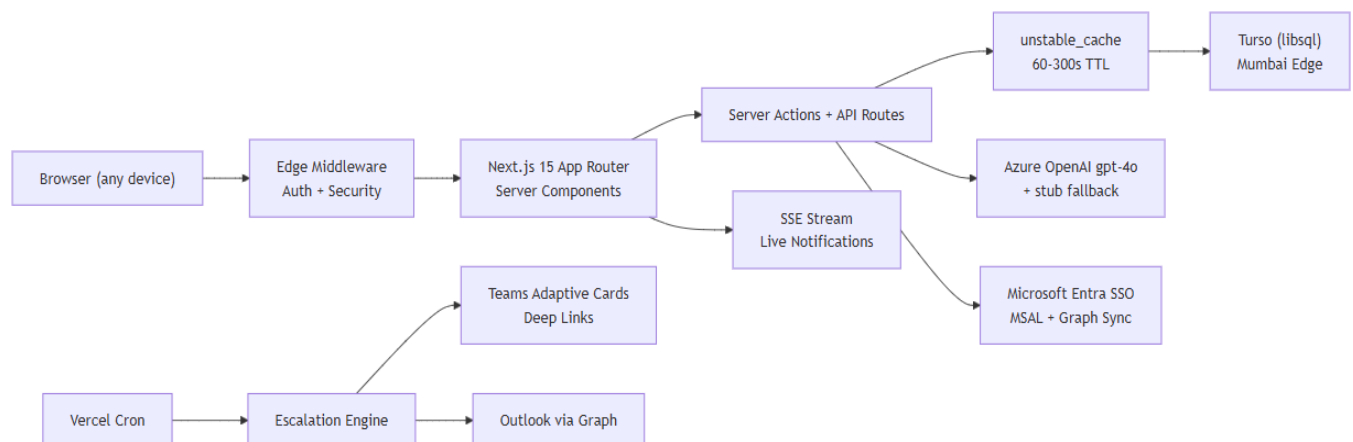
AtomicPulse — Hackathon Submission

AI-first Goal Setting & Tracking Portal | AtomQuest Hackathon 2026

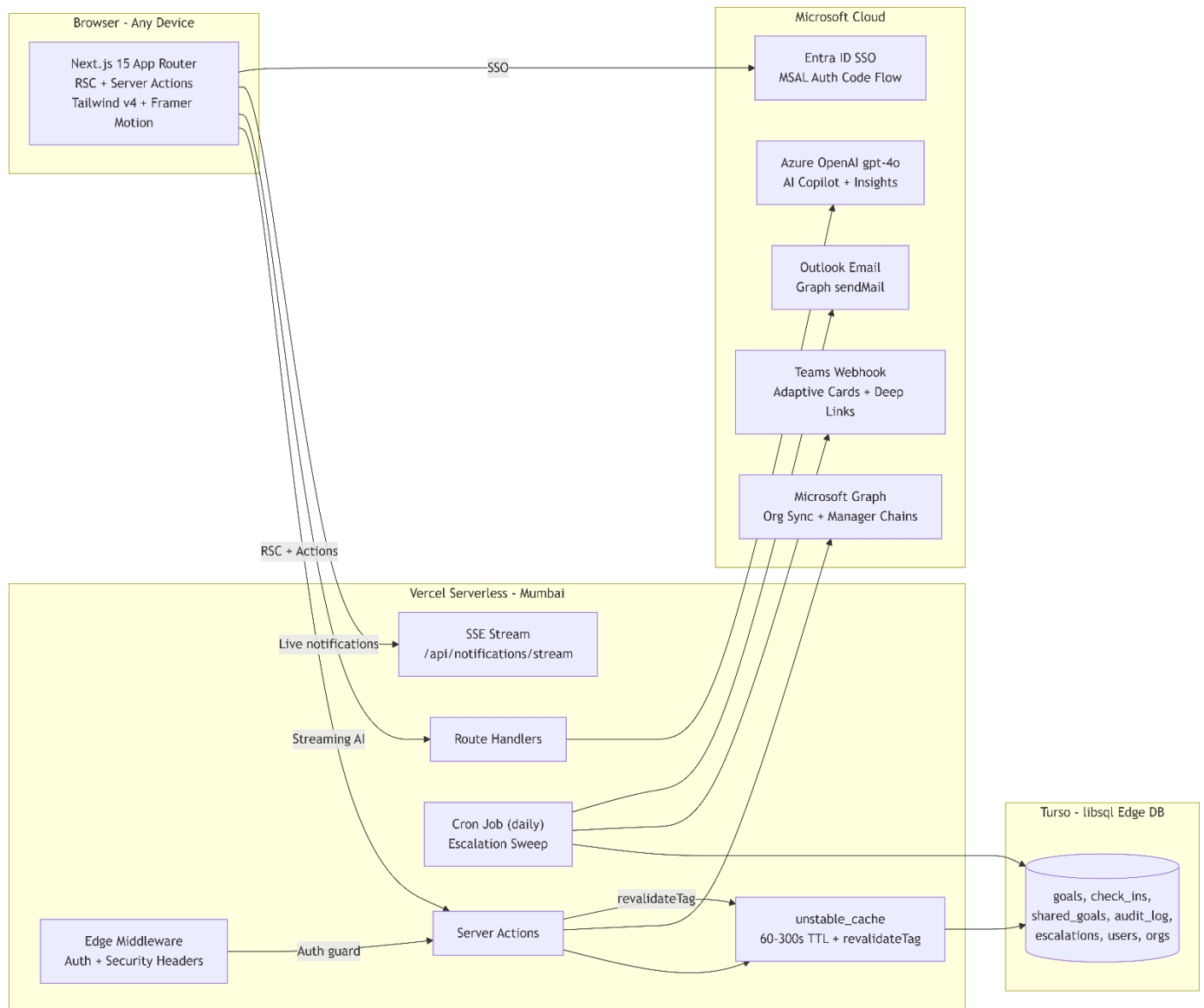
Submission Deliverables:

#	<u>Deliverable</u>	<u>Link</u>
1	Live Demo	atomic-pulse-aj5.vercel.app
2	Source Code	github.com/Adit-Jain-srm/AtomicPulse
3	Architecture Diagram	See below + AtomicPulse/tree/main/specs
4	Login Credentials	Demo Mode + Microsoft SSO — no passwords needed

Architecture:



High-level Architecture:



Key Technical Differentiators:

1. Tri-mode AI — stub (offline/free) → gateway (Vercel AI Gateway) → azure (direct OpenAI). Demo always works. Live AI generates real insights from actual goal data.
2. Real Microsoft 365 Integration — Not mocked. MSAL SSO with user upsert, Graph org sync with manager chain resolution, Teams Adaptive Cards with deep links, Outlook transactional email.
3. 152 Automated Tests — Every BRD rule tested at boundary conditions. Scoring formulas, validation rules, state machine transitions, edge cases, escalation triggers, analytics aggregation.
4. End-to-End Lifecycle Proof — Single Playwright test that walks: employee submit → manager approve → employee check-in — on one database without resets.
5. \$0 Hosting — Entire app on Vercel Hobby + Turso free tier. No Redis, no external auth provider, no notification SaaS. Framework-level caching with tag-based invalidation.
6. Production-Ready Security — Edge middleware auth guard, RBAC on every action, httpOnly signed cookies, security headers (nosniff, DENY framing, strict referrer), insert-only audit trail.

Section 5 BRD:

<u>Feature</u>	<u>Implementation Depth</u>
AI Copilot	7 skills (draft goals, score analysis, risk detection, quarter summary, KPI suggestions). Live Azure OpenAI with 25s timeout + deterministic stub fallback. Live insights on dashboard generated from actual goal data.
Microsoft Entra SSO	Real @azure/msal-node ConfidentialClientApplication. Auth code flow, token exchange, user upsert/update on login.
Graph Org Sync	Pages /v1.0/users, resolves manager chains, role mapping from AD group membership (GRAPH_ADMIN_GROUP_ID).

<u>Feature</u>	<u>Implementation Depth</u>
Teams Notifications	Adaptive Card 1.5 with deep links for every lifecycle event (submit, approve, return, check-in, escalation).
Outlook Email	Graph sendMail with HTML templates. Transactional emails for submissions, approvals, returns, reminders.
Escalation Engine	3 configurable triggers, threshold-day chain (owner→manager→skip-level→HR), deduplication, overlap prevention, daily cron.
Real-time Notifications	SSE stream polling every 8s, delta-based updates, live unread badge with dropdown panel.
Performance Analytics	QoQ trends (LineChart), heatmap (color-coded table), thrust distribution (BarChart), UoM mix (PieChart), manager effectiveness (BarChart).

About the Developer:

Adit Jain

GitHub	github.com/Adit-Jain-srm
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Resume	canva.link/Adit-Jain-CV